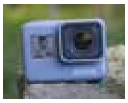

GoPro cameras everywhere
GoPro licenses its intellectual property with electronics maker Jabil to make third party lenses and cameras. **http://dgit.in/goprojabil**

YouTube simplifies livestreaming
YouTube adds dedicated live button on the website, as well as on Android and iOS apps. **http://dgit.in/Youtbelive**

Of **rows** and **columns**

Spreadsheets used in every office today started off as a paltry few rows and columns

Arnab Mukherjee | arnab@digit.in

Perhaps the most evident testament to the importance of office software right now is that this article probably wouldn't be written without it. Whether your software of choice is one of Microsoft's offerings or the more recent cloud based G Suite from Google, it is undeniable that office productivity software has pervaded every aspect of work life in practically any field of work today. What is packaged today as an overall productivity suite from most providers, had pretty disconnected and separate beginnings.

For the purpose of this article, we'll focus on one of the most significant type of office software, the spreadsheet. While initially conceived to deal with arithmetic and mathematical operations, today, spreadsheets are used for almost any application where tables and tabular lists are required. (Little known secret : a huge part of the work behind planning this magazine happens on spreadsheets). Most spreadsheet software today can deal with complex formulae, charts, graphs and macros, but they had a pretty humble beginning back in the 60s.

PAPER TALES

To put a date on the first spreadsheet would be to try and find the first time a human did their accounts. The term 'spreadsheet' was originally used to refer to bookkeeping ledgers - with the columns indicating categories for expenditure on top, the individual invoices listed on the left, and the intersection of the two holding the amount that was spent in that category for that invoice. This method was ubiquitous in the business world.

An electronic spreadsheet, as a concept, was first outlined by Richard Mattessich in a 1961 paper on "Budgeting Models and System Simulation". Subsequent works by Mattessich mainly dealt with adding or subtracting entire rows instead of single cells - still, as a concept. It was applied first using FORTRAN on the IBM 1130 in 1962.

Pre-electronic spreadsheet days

LANPAR or LANguage for Programming Arrays at Random, invented in 1969 by Remy Landau and Rene Pardo, was used for budgeting at Bell Canada, AT&T, and General Motors in mainframes and time-shared systems. The birth of the modern times of microcomputer-based electronic spreadsheet software came much later.

ON DISPLAY

Up until the early 1970s, electronic spreadsheets were mostly linear. Which means, that for every change made to the a value in the sheet, the entire sheet had to be recalculated again. Sitting in a lecture at Harvard Business School in the late 70s, Dan Bricklin realised that there had to be a better way to deal with the table structures than what the lecturer was doing - using a blackboard ruled with vertical and horizontal lines, where he had to erase and populate several entries in the table each time he found an error or change. Together with MIT acquaintance Bob Frankston, in the winter of 1978-79, the duo created VisiCalc for two months and founded the Software Arts company, putting their entire experience in working with report generators and electronic spreadsheets into use. Up until this point of time, report generators were not really user friendly, interactive, and couldn't run beyond mainframes or time sharing systems. VisiCalc changed all of that.

While the initial design by Bricklin contained a measly 5 columns and 20 rows, Frankston's code improved its speed, arithmetic and scrolling to a great extent and still ensured that